Zeju Li

GitHub | Google Scholar | Homepage | E-mail

Education

Zhejiang University Hangzhou, Zhejiang, China
B.Eng. in Computer Science and Technology

- College of Computer Science and Technology *Sep 2025 – Jun 2027 (expected)*
- Mixed Class, Chu Kochen Honors College *Sep 2023 – Jun 2025*

Research Interest

I'm passionate about developing truly intelligent robots capable of general-purpose autonomy in human-centric environments. My research centers on world models, robot foundation models, and agentic architectures that enable robots to generalize across tasks and learn efficiently through interaction. I am particularly interested in unifying of perception, locomotion, and dexterous manipulation within a single embodied agent as a step toward general intelligence grounded in the physical world.

Research Experience

Zhejiang University Hangzhou, Zhejiang, China
Research Assistant at State Key Lab of CAD&CG *Oct 2024 – Present*
Advised by Prof. Hao Chen and Prof. Chunhua Shen

Learning Robust Alignment from Critical Motion Segments
Mar 2026 – Jun 2026

- Co-led the study showing that fluent demonstrations underrepresent critical alignment and recovery states in fine-grained manipulation.
- Identified a supervision mismatch: critical alignment and recovery are compressed into brief motion segments and underrepresented during imitation learning.
- Proposed STAIR, a compact spatiotemporal interface that turns local motion into dense supervision and conditions vision-language-action policies on short-horizon motion.

Reconstructing, Generating, and Retargeting Hand-Object Interactions
Sep 2025 – Nov 2025

- Investigated how reconstructed human hand-object interactions can be transferred into simulation as dexterous-manipulation demonstrations.
- Built a real-to-sim pipeline that retargets MANO hand trajectories to a dexterous robotic hand and imports reconstructed object assets into Isaac Gym, validating AGILE's simulation readiness.

Compact State Representations for Efficient World Modeling and Action Learning
Jul 2025 – Sep 2025

- Co-developed the hypothesis that robot observations can be compressed into a highly compact token space in which motion emerges as differences between states.
- Led the study on bimanual manipulation data, examining whether severe compression can preserve the structure underlying coordinated actions.
- Showed through targeted representation probing that compact state differences retain linearly decodable action information, supporting a unified representation of state and motion.

Language-Guided Long-Horizon Mobile Manipulation

Jun 2025 – Aug 2025

- Contributed to a unified framework integrating hierarchical vision-language planning with whole-body control for long-horizon quadruped manipulation in open-world environments.
- Led the construction of its long-horizon benchmark, designing compositional tasks and evaluation protocols and establishing baseline performance across diverse indoor and outdoor scenarios.

Publications

* equal contributions, † corresponding author

[1] **Perfect Demo Makes Poor Teacher: Learning Robust Alignment from Critical Motion Segments**
Mingyu Liu*, Zeju Li*, Jiuhe Shu, Hanqing Wang, Yuhao Chao, Hao Chen†, Chunhua Shen†
CORL, 2026 [Paper] [Project Page]

[2] **StaMo: Unsupervised Learning of Generalizable Robot Motion from Compact State Representation**
Mingyu Liu*, Jiuhe Shu*, Hui Chen, Zeju Li, Canyu Zhao, Jiange Yang, Shenyuan Gao, Hao Chen†, Chunhua Shen†
CVPR, 2026 (Highlight Presentation, Top 3.1%) [Paper] [Project Page]

[3] **ODYSSEY: Open-World Quadrupeds Exploration and Manipulation for Long-Horizon Tasks**
Kaijun Wang*, Liqin Lu*, Mingyu Liu, Jianuo Jiang, Zeju Li, Bolin Zhang, Wancai Zheng, Xinyi Yu, Hao Chen†, Chunhua Shen†
AAAI, 2026 (Oral Presentation, Top 3.5%) [Paper] [Project Page]

[4] **AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation**
JinChuan Shi*, Binhong Ye*, Tao Liu, Junzhe He, Yangjinhui Xu, Xiaoyang Liu, Zeju Li, Hao Chen†, Chunhua Shen†
SIGGRAPH, 2026 (Conference Track) [Paper] [Project Page]

Skills

Programming Python (PyTorch), C/C++, LaTeX, SQL, Linux
Languages Chinese (Native), English (TOEFL 100/120, *Sep 2025*)